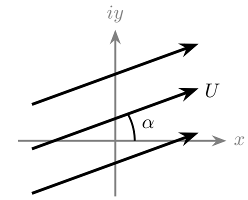
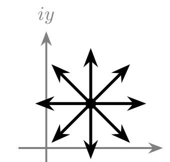
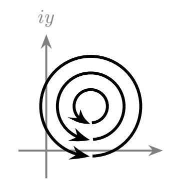
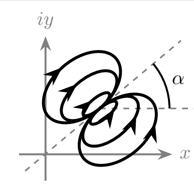
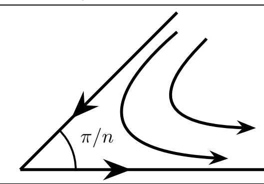
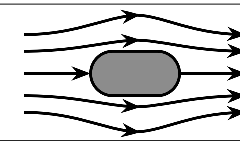
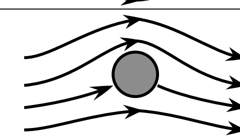
A Brief Handbook of Elementary Ideal Flows

Elementary flow	**Complex potential**, $F = \phi + i\psi$	**Potential**, $\phi(x, y)$ or $\phi(r, \theta)$	**Streamfunction**, $\psi(x, y)$ or $\psi(r, \theta)$	**Sketch of streamlines**
1. Uniform flow at angle α to the horizontal	Az with $A = Ue^{-i\alpha}$ U is real but A is complex	$U(x\cos\alpha + y\sin\alpha)$ or $Ur\cos(\theta-\alpha)$	$U(y\cos\alpha - x\sin\alpha)$ or $Ur\sin(\theta-\alpha)$	
2. Source ($Q > 0$; for sink replace Q by $-Q$) located at $z_0 = x_0 + iy_0$	$\dfrac{Q}{2\pi}\log(z - z_0)$	$\dfrac{Q}{2\pi}\ln\sqrt{(x-x_0)^2+(y-y_0)^2}$ or, for $z_0 = 0$, $\dfrac{Q}{2\pi}\ln r$	$\dfrac{Q}{2\pi}\tan^{-1}\left(\dfrac{y-y_0}{x-x_0}\right)$ or, for $z_0 = 0$, $\dfrac{Q}{2\pi}\theta$	
3. Vortex (counterclockwise for $\Gamma > 0$) located at $z_0 = x_0 + iy_0$	$\dfrac{-i\Gamma}{2\pi}\log(z - z_0)$	$\dfrac{\Gamma}{2\pi}\tan^{-1}\left(\dfrac{y-y_0}{x-x_0}\right)$ or, for $z_0 = 0$, $\dfrac{\Gamma}{2\pi}\theta$	$-\dfrac{\Gamma}{2\pi}\ln\sqrt{(x-x_0)^2+(y-y_0)^2}$ or, for $z_0 = 0$, $-\dfrac{\Gamma}{2\pi}\ln r$	
4. Doublet located at $z_0 = x_0 + iy_0$ and rotated by angle α	$\dfrac{me^{i\alpha}}{z - z_0}$	$\dfrac{m\cos\left(\alpha - \tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)\right)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$ or, for $z_0 = 0$, $\dfrac{m\cos(\alpha-\theta)}{r}$	$\dfrac{m\sin\left(\alpha - \tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)\right)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$ or, for $z_0 = 0$, $\dfrac{m\sin(\alpha-\theta)}{r}$	
5. Flow in a corner of angle π/n ($n > 1/2$) $1 < n < 2$: flows past wedges of angle $2\pi\frac{n-1}{n}$ $n = 2$: stagnation-point flow onto a horizontal plate	Az^n	[inconventient to use Cartesian coords] or $Ur^n\cos(n\theta)$	[inconventient to use Cartesian coords] or $Ur^n\sin(n\theta)$	
6. Flow past a Rankine body (a is a real number/horizontal shift) 1$\{\alpha = 0\}$ + 2$\{z_0 = -a\}$ − 2$\{z_0 = a\}$	$Uz + \dfrac{Q}{2\pi}\log\left(\dfrac{z+a}{z-a}\right)$	$Ux + \dfrac{Q}{2\pi}\ln\sqrt{\dfrac{(x+a)^2+y^2}{(x-a)^2+y^2}}$ [inconventient to use polar coords]	$Uy + \dfrac{Q}{2\pi}\left[\tan^{-1}\left(\dfrac{y}{x+a}\right) - \tan^{-1}\left(\dfrac{y}{x-a}\right)\right]$ [inconventient to use polar coords]	
7. Flow past a cylinder with circulation 1$\{\alpha = 0\}$ + 4$\{\alpha = 0, m = UR^2, z_0 = 0\}$ − 3$\{z_0 = 0\}$	$U\left(z + \dfrac{R^2}{z}\right) + \dfrac{i\Gamma}{2\pi}\log z$	$U\left(1 + \dfrac{R^2}{x^2+y^2}\right)x - \dfrac{\Gamma}{2\pi}\tan^{-1}\left(\frac{y}{x}\right)$ $U\left(1 + \dfrac{R^2}{r^2}\right)r\cos\theta - \dfrac{\Gamma}{2\pi}\theta$	$U\left(1 - \dfrac{R^2}{x^2+y^2}\right)y + \dfrac{\Gamma}{2\pi}\ln\sqrt{x^2+y^2}$ $U\left(1 - \dfrac{R^2}{r^2}\right)r\sin\theta + \dfrac{\Gamma}{2\pi}\ln r$	

Inspired by Tables 12.1 and 12.2 of Granger's *Fluid Mechanics* (Dover Publications, 2012). Flow 1–5 are the "building blocks" and the rest are superposition examples.